Direct lighting on meso-structured surfaces with area light sources

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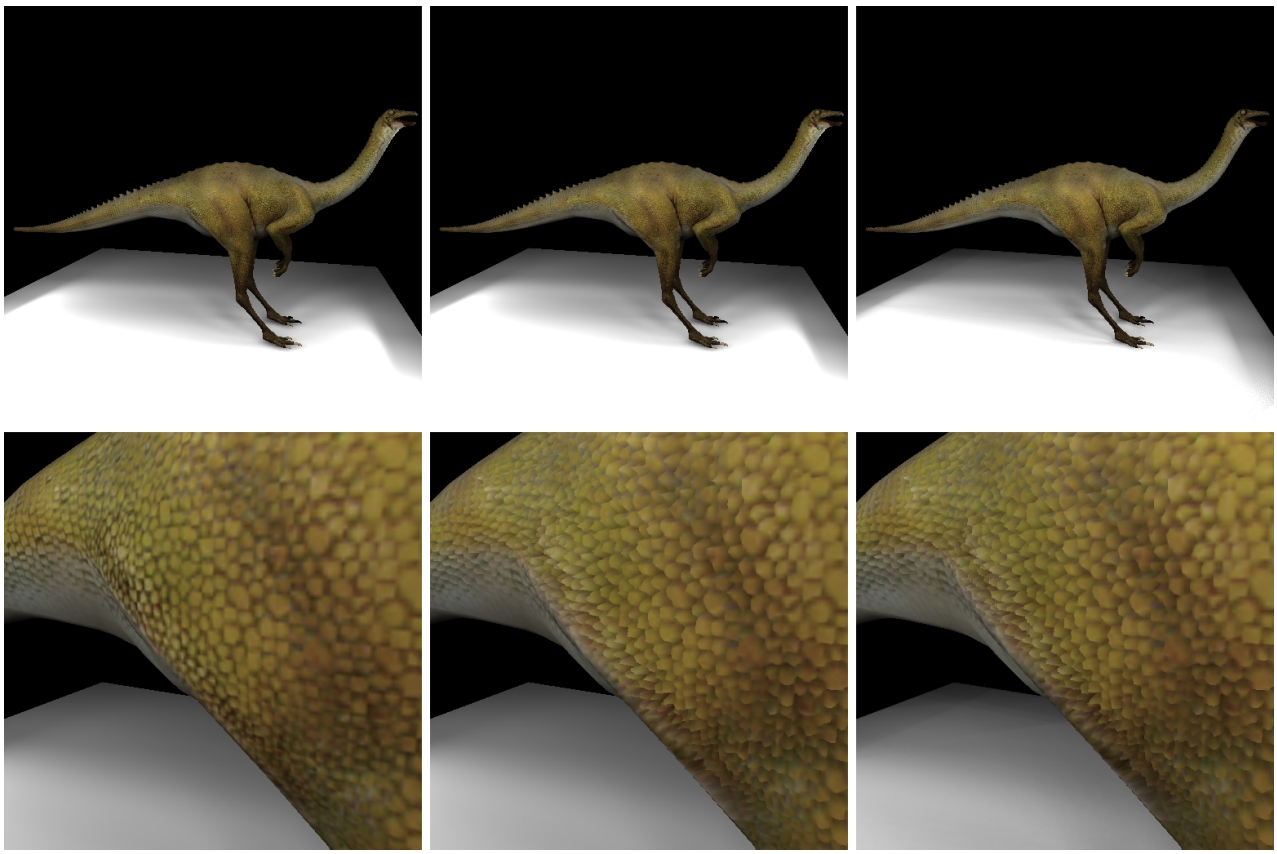


Figure 1: *Direct lighting on smooth textured surfaces rendered with radiosity method (left column), direct lighting on meso-structured surfaces rendered with our method (middle column) and with shadow mapping&relief texture mapping method (SM+RTM, right column) for comparison. The output image resolution is 512*512. The number of directional radiosity components in our method is 32. From the images, we can see the direct lighting rendered with our method is very similar to that of SM+RTM. The frame rate of our method vs. that of SM+RTM is 75.9fps vs. 10.0fps for the Dinosaur scene, which indicates our method has time performance advantage over SM+RTM for rendering direct lighting on meso- structured surfaces.*

Abstract

We present a novel direct lighting rendering method for static scenes with meso-structured surfaces and multiple area light sources. We use coarse geometry models with textures bound on them as inputs. The textures contain not only color maps, but also height field maps to represent the small concave and convex details for meso-structured surfaces. We explore traditional radiosity process as the main frame of our algorithm, and extend radiosity to directional radiosity for approximating light transportation on meso-structured surfaces. In pre-processing step, first, the form factors between the triangles of coarse models and the patches of area light sources are pre-computed. Then the directional radiosity of each triangle is computed by gathering irradiance according to directions sampled uniformly on the hemi-sphere of them. At last, directional radiosity of a vertex is estimated according to the directional radiosity of triangles passing through this vertex. For each frame, we combine relief texture mapping with directional radiosity to shade meso-structured surface. The experiment results show that our method has very similar quality and better time performance for rendering the scenes with meso-structured surfaces and multiple area light sources compared with SM+RTM method.

CR Categories: I.3.3 [Computer Graphics]: Three-Dimensional Graphics and Realism—Display Algorithms I.3.7 [Computer

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Graphics]: Three-Dimensional Graphics and Realism—Radiosity;

Keywords: direct lighting, meso-structured surfaces, directional radiosity

1 Introduction

There are many objects with rough surfaces in the real world. Light propagations between those objects are affected by meso-structures. Thus, meso-structured surfaces exhibit different appearance from that of smooth ones. Simulating the interactions between meso-structure details and light rays can help us to render the scene more realistic.

For the objects whose meso-structures are modeled with actual geometry details, i.e. the higher resolution model with lots of vertices and triangles, monte-carlo ray tracing [Lafortune 1996; Jensen et al. 2003; Veach and Guibas 1997] and traditional radiosity method [Goral et al. 1984; Greenberg et al. 1986] can be used to render high quality direct lighting effect. However, quantities of ray-triangle intersections in Monte-carlo ray tracing and a large number of form factor determinations in radiosity method become the bottleneck of computation, which lead unacceptable waiting time for generating result images. Especially for some interactive applications such as tools for model preview, virtual surgery for organs or tissues and image composite AR applications, the developers always want to demonstrate the high quality images with many small geometry details in real-time. These applications proposed challenges for the classical methods.

In order to accelerate direct lighting computation for meso-structured surfaces, relief texture mapping [Oliveira et al. 2000; Policarpo and Oliveira 2006; Policarpo et al. 2005]was proposed for real-time applications. Relief texture mapping has very good time performance for the scene with point light sources. However, for the scene with area light sources, we need to approximate the area light source with a set of point light sources. For each point light source, relief texture mapping is used for each sample on meso-structured surfaces. Thus, the time cost is in proportion to the number of point light sources sampled from the area light sources. Moreover, relief texture mapping need to be combined with shadow mapping to render the shadows of the coarse model without meso-structure details, which also adds extra time costs for rendering direct lighting.

Our method is designed to render direct lighting for static scenes with meso-structured surfaces and multiple area light sources. We extend radiosity to directional radiosity for approximating light transportation on meso-structured surfaces, and present an algorithm to compute directional radiosity in the gathering process. We use coarse geometry models textured with height fields to represent geometry details of meso-structures as inputs, and explore traditional radiosity method as the main frame of the process. A quantity of form factor calculation for meso-structured surfaces is avoided by directional radiosity computation. Our method only pre-computes the form factors between the triangles of coarse models and the patches of area light sources. Thus, the pre-computation time of our method is as the same level as that consumed for rendering the scene without meso-structured surfaces by using traditional radiosity method. A rendering algorithm based on directional radiosity clustering is proposed to render direct lighting for meso-structured surfaces. For static scenes, multiple shadow checking based on the pre-computed shadow map is avoided in our method, which also brings some performance benefits for scenes with complex area light sources when viewpoint is moving. The experiment results show that our method can approximate direct lighting for the scene with meso-structured surfaces in high quality and be efficient for rendering scenes with multiple area light sources(see Figure 1 and accompanying video). Our method sustains above 65fps and is about 7$\sim$10 times faster than SM+RTM for all four static scenes.

2 Prior Work

2.1 Radiosity method

Classical radiosity method [Goral et al. 1984; Greenberg et al. 1986] which describes an equilibrium energy balance within an enclosure is a technique to render global illumination of diffuse environments. In the method, the scene is discretized into surface patches. An important parameter of this method is the form-factor matrix which represents the fraction of energy leaving one surface patch and arriving directly at another. The form-factor matrix needs to be pre-computed, which is expensive for large scenes. Different strategies have been proposed to accelerate the original algorithm, such as progressive radiosity [Cohen et al. 1988], instant radiosity [Keller 1997], hierarchical radiosity [Hanrahan et al. 1991], hierarchical image-space radiosity [Nichols et al. 2009] and hemi-cube method [Cohen and Greenberg 1985] etc. In progressive radiosity method, patches are processed in sorted order according to their energy contribution to environment. The progressive method can calculate the form-factors on-the-fly to eliminate the O(n^2) storage and startup costs. Hemi-cube method is an efficient algorithm to compute form factors in general complex environments. In this paper, we explore classical radiosity method as the main frame, then we extend radiosity to directional radiosity for approximating light transportation on meso-structures. The hemi-cube method is used in the shooting process. We use directional radiosity method to gather energies.

2.2 Lighting on meso-structured surface

Objects with meso-structured surfaces have more details than smooth ones. Many methods, such as bidirectional texture function (BTF) [Dana et al. 1999; Tong et al. 2002; Suykens et al. 2003], bi-scale radiance transfer (BRT) [Sloan et al. 2003], shell texture function(STF) [Chen et al. 2004] and shell radiance texture function (SRTF) [Song et al. 2005] have been researched deeply for objects with meso-structured surfaces. But these algorithms require texture synthesis and need much storage.

The concavo-convex effects for the objects with meso-structured surfaces can be rendered by using some texture mapping methods, such as bump mapping [Blinn 1978], displacement mapping [Cook 1984], parallax mapping [Kaneko et al. 2001], horizon mapping [Max 1988; Sloan and Cohen 2000], relief texture mapping and some improved methods, such as normal mapping [Heidrich and Seidel 1999], parallax occlusion mapping [Tatarchuk 2006]. An warping-based technique called relief texture mapping was presented by Oliveira et al [Oliveira et al. 2000], which can capture occlusion and silhouettes, but shading and self-shadowing are absent. The method used image warping techniques and textures enhanced depth of per-texel to create the lighting effect of complex geometric details when mapped onto flat polygons. Policarpo et al. [Policarpo et al. 2005] presented relief texture mapping to render a textured mesh. The method transforms the point and corresponding view direction to view space for each pixel which contains a height. The view ray is traversed until it hits a height field, and binary search is used to increase the accuracy of intersection. Oliveira and Policarpo [Oliveira and Policarpo 2005] tried to render silhouettes by locally approximating surface with a piecewise quadric representation . In 2006, Policarpo [Policarpo and Oliveira 2006] presented an extend method that renders multiple image layers instead of height field. Snyder and Nowrouzezahrai [Snydre

and Nowrouzezahrai 2008] presented a spherical harmonics-based approach for soft self-shadowing on dynamic height fields. In this method, Snyder and Nowrouzezahrai have approximate the horizon blocking of environment light by visibility wedges and use pretabulated SH-projections for them to determine the overall visibility, but the method is restrict to the special case of dynamic height fields. In 2009, Nowrouzezahrai and Snyder [Nowrouzezahrai and Snyder 2009] extended the method to indirect illumination and accelerated the method. Sun et al [Sun et al. 2011] presented an all-frequency BRT method which is based on biclustering. This method can support high-frequency biscale lighting effects in real time, but it only focuses on compressing the macro-scale radiance transfer data. In this paper, relief texture mapping is combined with directional radiosity to render the direct lighting for meso-structured surfaces.

2.3 View-dependent displacement mapping(VDM)

Wang et al. [Wang et al. 2003] presented a method which precomputed the distances from each displaced point to a reference surface along many sampling viewing directions, resulting a five-dimensional function called a view-dependent map(VDM) that can be queried at rendering time. Data decomposition and compression need to be done by singular-value decomposition(SVD), because of high-dimensional maps and the large sizes. The method can work in real time using a GPU and produce nice results. The method can lead to significant texture distortions and only be suitable for closed surfaces. In order to solve above problem , another five-dimensional representation called generalized displacement map(GDM) [Wang et al. 2004] was introduced by Wang et al. GDM is used to render non-height-field structures and also produces large databases that need to be decomposed and compressed. However, GDMs are more suitable for tiling and renderings from a certain distance. The above two methods only are suitable for scene with point light source, but we want to render the direct lighting on meso-structured surfaces with multiple area light sources.

2.4 SM+RTM

SM+RTM is a conventional method to approximate direct lighting on meso-structured surfaces with area light sources, which is combination of shadow map and relief texture mapping. The SM+RTM method takes coarse geometry models with textures bound on them as input, which is appropriate to be contrast method.

SM+RTM renders direct lighting on meso-structured triangles as follows:

1. Pre-processing:
   1) Shadow map method is used to precompute depth map for each small light patch
2. Online processing:
   1) Observe scene from viewpoint and get some texture which deferred shading need
   2) Relief texture mapping in the view space for each light patch

In step 1) of pre-processing the area light sources are divided into small light patches. Shadow map method is used to decide visibility between triangles and each small light patch. Deferred shading is used in online processing. At last, relief texture mapping is execute once for each small light patch.

3 Directional Radiosity

In traditional radiosity method, for a given patch *i*, its radiosity B_i can be computed as following:

$$B_i = E_i + \rho_i \sum_{j=1}^{n} B_j F_{ij} \tag{1}$$

where B_j is radiosity of patch *j*, E_i is emission of patch *i*, F_{ij} is a form-factor of *i* and *j*(fraction of energy leaving patch *i* which arrives at patch *j*) and ρ_i is reflectivity of patch *i*.

Traditional radiosity method assumes all patches of geometry models are ideal diffuse, thus the radiosity B_i of all directions are regarded as the same intensity. However, if patch *i* has some meso-structures, it may exhibit different radiosity along the different directions because of concavo-convex geometry details. Thus, directional radiosity for meso-structured patch *i* can be defined as $dRad_i = \{D_i, B_i\}$: $D_i = \{d_i^1, d_i^2, d_i^3 \dots d_i^n\}$ is the set of n directions in the hemi-sphere of patch *i*; $B_i = \{b_i^1, b_i^2, b_i^3 \dots b_i^n\}$ is the set of radiosity on n directions, and we call $b_i^1, b_i^2, b_i^3, \dots$ as directional radiosity components of patch *i*. Figure 2 shows meso-structure surfaced patch i with directional radiosity components of different directions in $i's$ hemi-sphere.

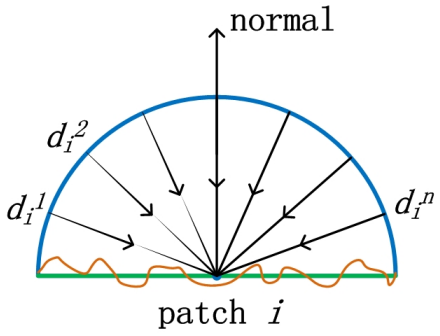


Figure 2: *Directional radiosity for meso-structured patch.*

4 Algorithm

4.1 Algorithm overview

Consider a 3D scene that contains both smooth and meso-structured surfaces. Smooth surfaces are modeled as regular triangles, while meso-structured surfaces are modeled as triangles bound with u-v coordinates of height maps to represent their geometry details. The scene also contains multiple area light sources. We use S to represent the 3D scene, T_i to represent the triangle of meso-structured surface, V_i^j to represent the vertex of T_i, L_k to represent light source. Direct lighting on smooth surfaces is computed and rendered with traditional radiosity method. Our algorithm renders direct lighting on meso-structured triangles as follows:

1. Pre-processing:
   1) Form factor calculation between each T_i and each small patches of each L_k
   2) Compute directional radiosity for each T_i
   3) Directional radiosity interpolation for each V_i^j of each T_i

2. Online processing:

   1) Relief texture mapping based on directional radiosity of V_i^j

In step 1) of pre-processing, the area light sources are divided into small light patches. Then form factors between these small light patches and triangles T_i are calculated with the traditional radiosity method. In step 2), a hemi-sphere is build according to the normal of T_i, n directions are sampled uniformly on the hemi-sphere to gather irradiance in different directions. After this, the directional radiosity components of n directions for T_i are computed, which is discussed in section 4.2. In order to get smooth shading, the directional radiosity of patches are interpolated to the vertex in step 3), and the detail of interpolation will be introduced in section 4.3. After pre-processing, relief texture mapping based on directional radiosity is proposed in section 4.4 to render the scene in real-time when the viewpoint is moving.

4.2 Directional radiosity component computation

Since the goal of our paper is to calculate direct lighting, only the radiosity from area light sources to geometry patches should be considered. For each meso-structured patch, the directional radiosity component of a give direction n_i is estimated according to radiosity in the similar directions of small light patches.

Figure 3 shows a 2D example of the radiosity gathering process. *L* is an area light source and *i* is a meso-structured patch, which is represented as a triangle in the 3D scene.We divide *L* into four small light patches l_1, l_2, l_3 and l_4.For each small light patch, a hemi-cube is constructed. Then the meso-structured patch is projected on the hemi-cube constructed. In Figure 3, i is projected on the hemi-cube of l_2, and i's projection covers on the cell a and b of l_2's hemi-cube. We get two directions V_2^a and V_2^b when the center of l_2 and the centers of cell *a* and *b* are connected. The radiance of l_2 is accumulated to the irradiance of patch *i* in direction d_i^5 and d_i^6, which are most similar to V_2^a and V_2^b.

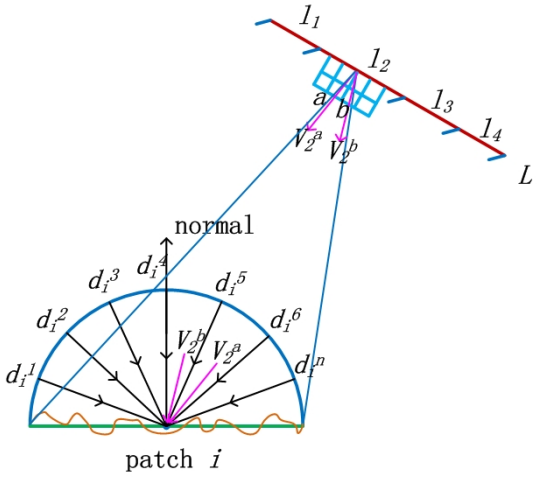


Figure 3: *Directional radiosity component computation for meso-structured patch.*

4.3 Vertex directional radiosity estimation

In order to render scenes smoothly, traditional radiosity method transfers radiosity of patches to vertexes. For each vertex, all neighbor patches are taken into account for a weighted average of radiosity. Then phong shading is used to guarantee smooth shaded images. However, the directional radiosity for patch contains radiosity components on several directions, the radiosity estimation for vertex in traditional method cannot be used. We extend this estimation to generate the directional radiosity for vertex. First, normal of vertex is computed by averaging surface normals of its neighbor patches. Second, a hemisphere is constructed for each vertex, which is taken the vertex normal as the main axis. N directions are sampled from the hemisphere uniformly, and are used as directions of directional radiosity for the vertex. At last, the radiosity of neighbor patch at a given direction is accumulated to the radiosity component on the most similar direction of the vertex.

Figure 4 shows a 2D example of our directional radiosity estimation for vertex. *i* and *j* are the neighbor patches of vertex *v*, the normal of *v* is averaged with the normals of *i* and *j*. N directions of directional radiosity for *v* are constructed as the same way as those of patches. For each direction *d* of both *i* and *j*, a most similar direction d_v is found. The value of directional radiosity component on *d* of patch is accumulated to that on d_v of vertex with linear interpolation. For the example in figure 4, d_v^4 of *v* is the similar direction of both d_i^6 and d_j^2. Thus, the value of direction radiosity component on d_v^4 is linear interpolated with the component values on d_i^6 for *i* and d_j^2 for *j*.

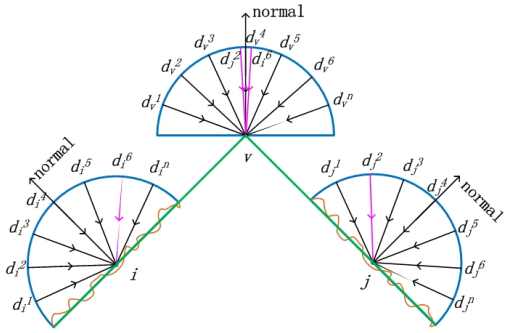


Figure 4: *directional radiosity on vertex.*

4.4 Directional radiosity based relief texture mapping

We extend relief texture mapping(RTM) with directional radiosity to shade meso-structured surface on-line. Algorithm 1 gives the steps of DR-RTM. *Step a* and *step b* are as same as regular RTM. In *step c* of regular RTM, the visibility between the real intersection *r* on meso-structured surface to the point light source is determined by intersecting the ray to the relief map, and phong model is used to compute the lighting of *r*. While our DR-RTM processes area light source lighting on meso-structured surface with directional radiosity, radiosity in multiple directions are considered to shade *r* in *step c*. We illustrate the details of *step c* in Algorithm 2.

Algorithm 2 takes $p(i,j)$, *r*, the mesh of scene and the directional radiosity of vertexes as input, and outputs the color of *r* at $p(i,j)$. The directional radiosity of vertex is transferred to that of pixel $p(i,j)$ with the consideration of $p(i,j)'s$ normal and position where it locates inside the corresponding triangle. In each direction, first, we compute its radiosity component for $p(i,j)$ with Algorithm 3, and show how to find the most similar directions of the neighbor vertexes for the directions at $p(i,j)$. Then, the visibility in this direction is determined with the relief map, which is as the same as regular relief texture mapping method. At last, we accumulate $r's$ color

Algorithm 1 DR-RTM

Input: coarse mesh of scene, hight field maps and normal maps
Output: effect of direct lighting and sense of Concavo-Convex for this scene

Render coarse mesh of scene from the output viewpoint;
for each pixel $p(i,j) \in$ the output image **do**
    step a: Find real occlusion point r in relief map;
    step b: Get the color c and normal of r;
    step c: Directional radiosity based shading r at $p(i,j)$;
end for

Algorithm 2 Directional radiosity based shading for r

Input: $p(i,j)$, r and the mesh of scene
Output: color at $p(i,j)$

Construct n directions in hemisphere of $p(i,j)$ with consider of $p(i,j)'s$ normal;
$color \leftarrow 0$;
Get triangle t that contains $p(i,j)$;
Get three weight values w[3] of $t's$ vertexes for $p(i,j)$;
for a given direction $d \in$ all directions **do**
    $rad \leftarrow$ Compute Radiosity component in this direction for $p(i,j)$;
    Compute the visibility on this direction with the relief map
    if visible **then**
        $color \leftarrow color$ + direction k . the normal of r * rad * c ;
    end if
end for
Shade r with color at $p(i,j)$.

with the radiosity component of $p(i,j)$ if there is not occlusion in this direction. After the radiosity components in all directions for $p(i,j)$ are processed, we shade r with the color accumulated.

Algorithm 3 Compute radiosity component in direction d for $p(i,j)$

Input: directional radiosity of vertexes, three weight values *w[0]*, *w[1]*, *w[2]* and triangle t that contains $p(i,j)$
Output: radiosity component in direction d for $p(i,j)$

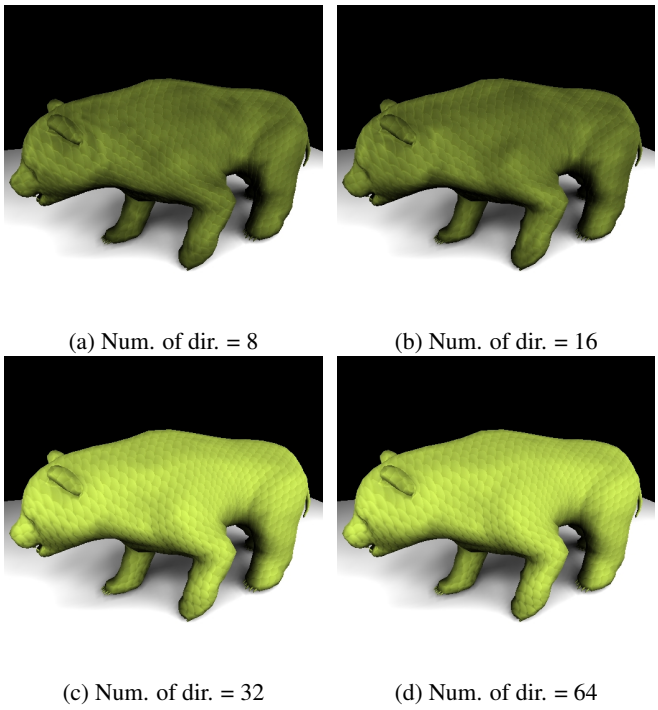
$rad = 0$
for each vertex $k \in$ three vertexes of t **do**
    $g \leftarrow$ find most similar direction of vertex
    $Rad \leftarrow$ get the value of radiosity component in direction of g
    $rad \leftarrow rad + w[k] * Rad$
end for
return rad

5 Results and Discussion

We tested our method on four static scenes with meso-structured surfaces. The *Bear* scene has 30.5k triangles. The *Torus* scene has 23.8k triangles. The *Dinosaur* scene has 66.1k triangles. The *Basket* scene has 50.9k triangles. In this paper, each area light source is divided into 32 patches. In the method of SM+RTM, we approximate each patch with one point light source.

5.1 Quality (compared to SM+RTM)

As shown in Figure 1, Figure 5 and in the accompanying video, our method produces quality direct light effects for the scene with meso-structured surfaces under area light sources. The experiment



(a) Num. of dir. = 8 (b) Num. of dir. = 16

(c) Num. of dir. = 32 (d) Num. of dir. = 64

Figure 6: *Direct lighting effects on meso-structure surfaces with various number of directional radiosity components.*

results shows there is very little difference between direct lighting rendered with our method and direct lighting with SM+RTM in the visual. As we explore traditional radiosity as the main frame, our method inherits the features of smooth from traditional radiosity which is visible in the shadow region of scene.

The direct lighting on meso-structured surfaces rendered with the various number of directional radiosity components are shown for the *Bear* in Figure 6. From the figure, we can see the various num of directional radiosity components has an influence on the brightness of meso-structures, but not the concave and convex sense of scene. As an incident ray must be merged into most similar directional radiosity component, the contribution to brightness of scene can become larger or smaller which will converge with large quantity of directional radiosity components. We can see that the difference between the effect with 32 directional radiosity components and that with 64 directional radiosity components is very small. In our test, 32 is the apposite number of directional radiosity components.

Figure 7 gives the comparison of direct lighting effects on meso-structure surfaces for the *Bear* scene by dividing area light source into patches with different size. It can be seen that various patch size of area lighting source can result in different direct lighting effects. The form-factor between area lighting source patch and each triangle is related to the triangle center, triangle normal, patch center and patch normal etc, other parameters apart from patch center are same for various area lighting source patch size. From above, we can see the deviation of form-factor deciding becomes larger when area lighting source patch size increases. By the comparison, we can find the direct lighting effect of dividing each area light source into 4*4*2 patches is very similar to that of dividing each area light source into 5*5*2 patches. In the following sections, It can be seen that decreasing light patch size will result in frame rate dropping. So, in this paper ,we divide each area light source into 4*4*2 patches.

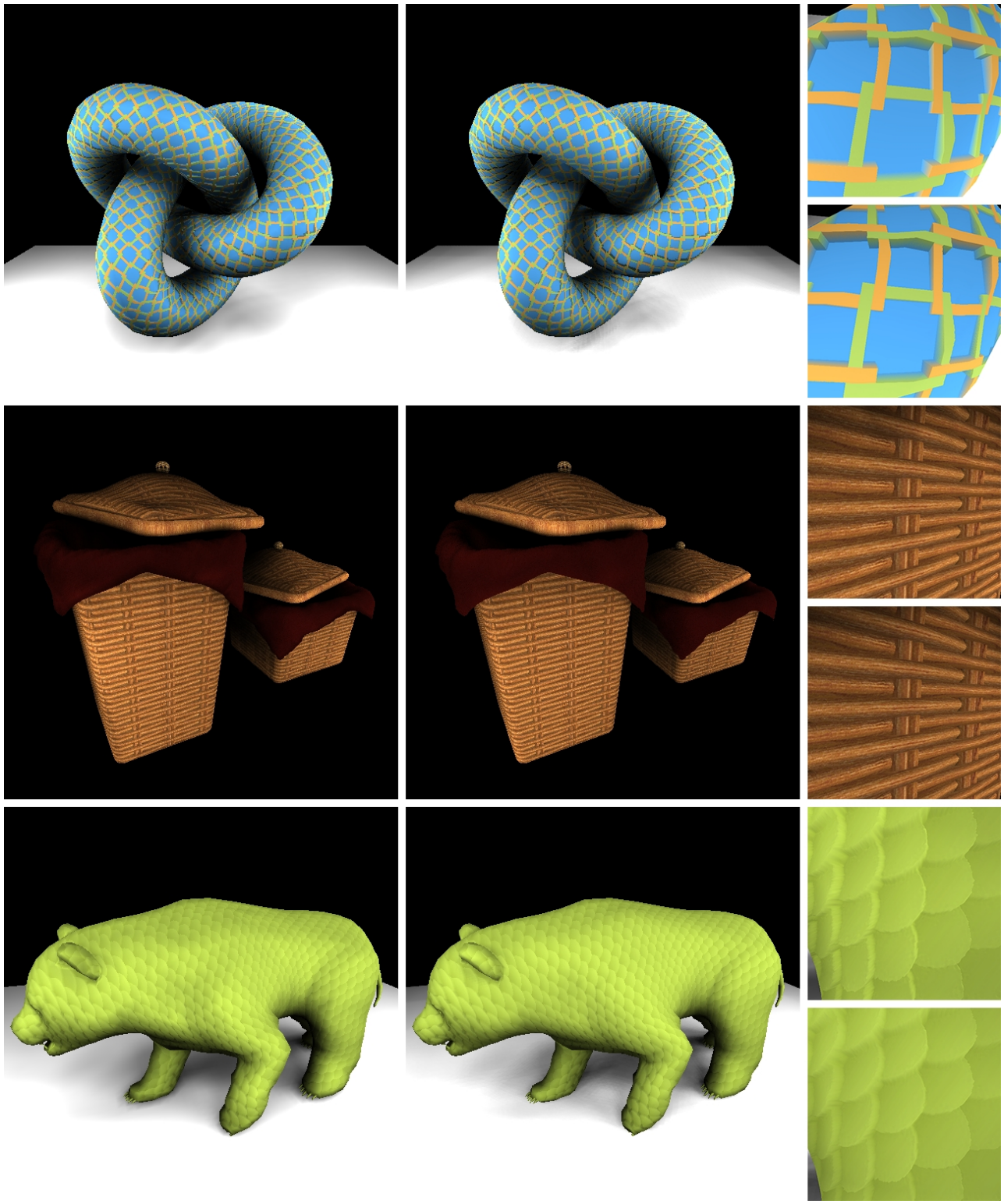
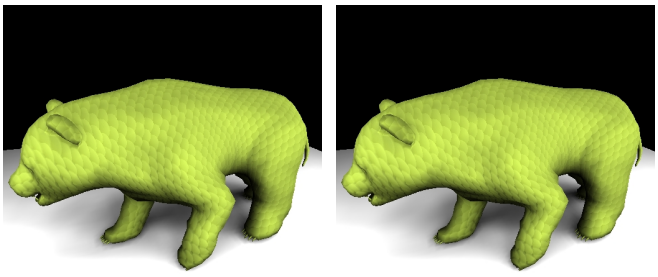
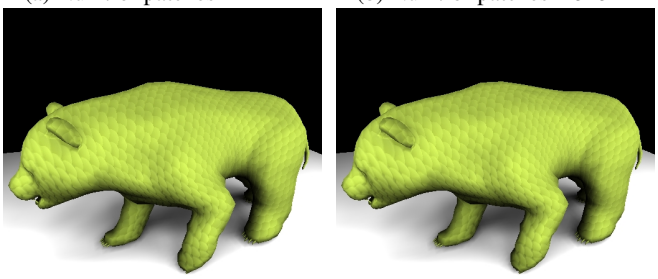


Figure 5: *Examples of direct lighting on meso-structured surfaces rendered with our method(left and top-right for each row) and with SM + RTM(middle and bottom-right for each row), for comparison. The frame rate for our method vs. SM+RTM is 92.7fps vs. 9.0fps for the Bear scene, 67.1fps vs. 9.9fps for the Torus scene, and 72.7fps vs. 9.9fps for the Baskets scene. The output image resolution is 512*512. Num of directional radiosity component in our method is 32.*



(a) Num. of patches = 2*2*2 (b) Num. of patches = 3*3*2



(c) Num. of patches = 4*4*2 (d) Num. of patches = 5*5*2

Figure 7: *The comparison of direct lighting effects on meso-structure surfaces for the Bear scene with various area light source patch size.In the test, we divide each area light source into different number of patches.*

5.2 Performances (compared to SM+RTM)

All performance measurements reported in this paper were recorded on a 3.4GHz Intel(R) Core(TM) i7-2600 CPU PC with 4 GB of RAM and a NVIDIA GeForce GTX 580 graphics card.

The performance of our method and the comparison to SM+RTM is given in Table 1. SM+RTM used precomputed shadow map and deferred shading. Performance was measured under 896 area light source patches. The pre-computation cost of our method is higher than that of SM+RTM, our method sustains above 65fps for all four static scenes and it is about 7∼10 times faster than SM+RTM. For our method, the pre-computation cost is related to the number of scene triangles and the proportion of meso-structured surfaces in the scene, The frame rate is only affected by the proportion of meso-structured surfaces on the screen. In the *Basket* scene, the pre-compute time is little, because of having a small amount of rasterized triangles for each area lighting source patch.

The rendering parameters that could affect performance are the number of area light sources (Table 2), the number of directions for directional radiosity (Table 3) and the output resolution (Table 4) . Tables 2, 3, 4 report performance for the *Bear* scene.

Table 2 gives the pre-computation time[s], frame rate [fps] and memory for various area light source number. Output resolution is 512*512, the number of directions is 32. From Table 2, we can include that as the number of area light sources increases, the advantage of our method becomes more obvious. Our method has a higher frame rate and lower memory requirements. We use directional radiosity to avoid some computation online by pre-computing illumination directions into similar directional radiosity components, that result in higher frame rate. Our method has lower memory cost, because SM+RTM has a shadow map for each small light patch. If we use 512*512 shadow maps, for 896 patches, the memory is 448, while our method is not sensitive to the number of area light source patches, so the memory will not increase. The number of direction for all vertexes influences the memory cost of our method, but we used very coarse geometry models, the number of vertexes is not large. From our experiments, 32 directions for each vertex are good enough to generate high quality images. Compared with its advantages, a little larger pre-computation cost is acceptable in some applications.

Our method extends radiosity to directional radiosity by using several directional radiosity components. Performance is affected by number of directions in directional radiosity. Table 3 gives the pre-computation cost and frame rate for the *Bear* Scene for various number of directions in directional radiosity. Output resolution is 512*512, the number of area light source patches is 896. In the stage of pre-computation, the number of directions only influences the cost of vertex directional radiosity estimation which is a very small part of pre-computation, so the pre-computation cost increase slowly as number of directions increasing. In the stage of online rendering, the number of directions has a great influence on online rendering.

In the stage of online rendering, performance depends on the output resolution: for a higher output image resolution, there are more pixels to compute the total effect of directional radiosity .Table 4 gives the pre-computation time and frame rate for various output resolution. The scene is *Bear* scene , number of directions is 32 and number of area light sources is 56. The output resolution has no influence on the pre-computation, but has a great influence on frame rate. This translates to slower frame rates for higher resolution.

6 Conclusions and Future Work

We have presented a novel approach to simulate the interactions between meso-structure details and light rays, which help us to render the scene more realistic. We extend radiosity to directional radiosity to render direct lighting for static scenes with meso-structures surfaces under multiple area light sources.

The results of the experiments show that our algorithm render the static scene quickly with a good quality. Compared with SM+RTM, our algorithm has a little more pre-processing time, but much less online processing time. With the increase of light sources, the advantage of our algorithm becomes more obvious.

Still, there are a few limitations in our method. As our method explores traditional radiosity method as the main frame, area lighting source and scene must be static. At the same time our method extends radiosity to directional radiosity which replaced irradiance with some directional radiosity components, so lighting source color can not be changed. In this paper, we used relief texture mapping to achieve the concave and convex sense of meso-structured surfaces, therefore, our method inherit the limitation that the rendering of meso-structured edges and contours can not be solved.

In terms of future work, we are interested in extending our method to dynamic scenes. We will use spherical harmonic basis to precompute propagation of light and abandon the high-frequency signal in the rendering process. For the scene with rough surfaces, the global illumination still need to compute the interactions between meso-structured details. We will extend our method to provide solutions to this case. Int this case, all scene patches should shoot the energy in the shooting process, we can explore progressive radiosity as the main frame.

Table 1: *Performance of our method compared to SM + RTM for meso-structured scenes*

Scene	Ours method		SM+RTM	
	Pre-comp.(s)	Online(fps)	Pre-comp.(s)	Online(fps)
Bear(30.5k)	23.0	92.7	4.0	9.0
Torus(23.8k)	22.7	67.1	3.8	9.9
Dinosaur(66.1k)	25.3	75.9	4.2	10.0
Baskets (50.9k)	38.1	72.7	1.0	9.9

Table 2: *Pre-computation time[s], frame rate [fps] and memory [M] for various area light source number*

Num of area light source patches	Ours method			SM+RTM		
	Pre-comp.(s)	Online(fps)	Memory(M)	Pre-comp.(s)	Online(fps)	Memory(M)
160	4.6	92.7	128	0.7	48.4	80
256	6.9	92.7	128	1.2	30.6	128
448	11.5	92.7	128	2.0	18.2	224
608	15.3	92.7	128	2.7	13.5	304
800	20.4	92.7	128	3.5	10.1	400
896	23.0	92.7	128	4.0	9.0	448

Table 3: *The pre-computation cost[s] and frame rate [fps] for various number of directions in Directional radiosity*

Num of directions	8		16		32		64	
	Pre-comp.(s)	Online(fps)	Pre-comp.(s)	Online(fps)	Pre-comp.(s)	Online(fps)	Pre-comp.(s)	Online(fps)
Bear	22.6	173.6	22.7	155.8	23.0	92.7	23.4	51.3

Table 4: *Pre-computation time[s] and frame rate [fps] for various output resolution*

Res.	256*256		512*512		1024*1024	
	Pre-comp.(s)	Online(fps)	Pre-comp.(s)	Online(fps)	Pre-comp.(s)	Online(fps)
Bear	23.0	164.7	23.0	92.7	23.0	28.2

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